Q1.

```
import seaborn as sns
import pandas as pd
import matplotlib.pyplot as plt
```

```
sns.set_style('whitegrid')
```

Q2

```
titanic = sns.load_dataset('titanic')
```

Q3

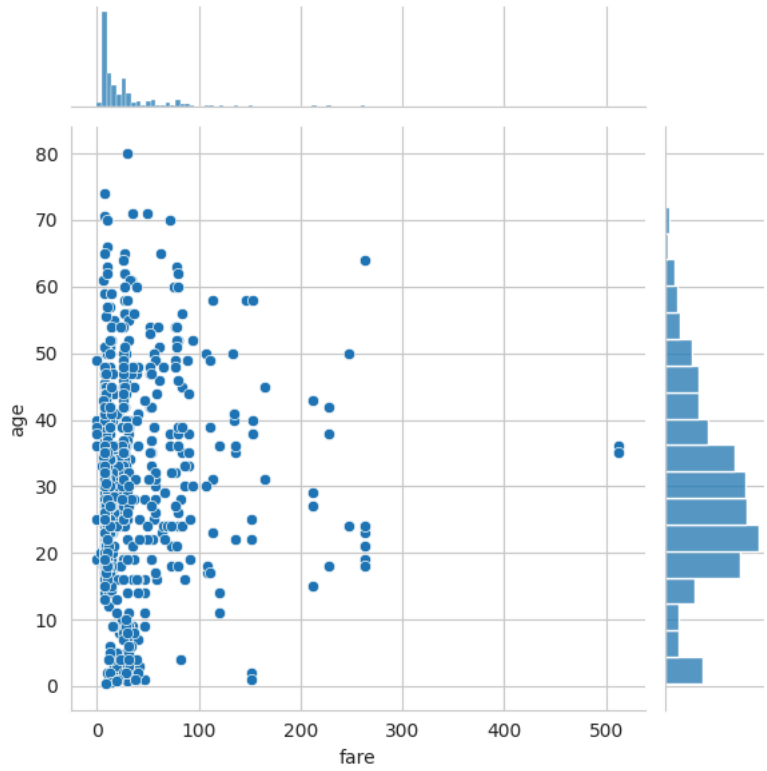
```
titanic.head()
```

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	who	adult_male	deck	embark_town	alive	alone
0	0	3	male	22.0	1	0	7.2500	S	Third	man	True	NaN	Southampton	no	False
1	1	1	female	38.0	1	0	71.2833	C	First	woman	False	C	Cherbourg	yes	False
2	1	3	female	26.0	0	0	7.9250	S	Third	woman	False	NaN	Southampton	yes	True
3	1	1	female	35.0	1	0	53.1000	S	First	woman	False	C	Southampton	yes	False
4	0	3	male	35.0	0	0	8.0500	S	Third	man	True	NaN	Southampton	no	True

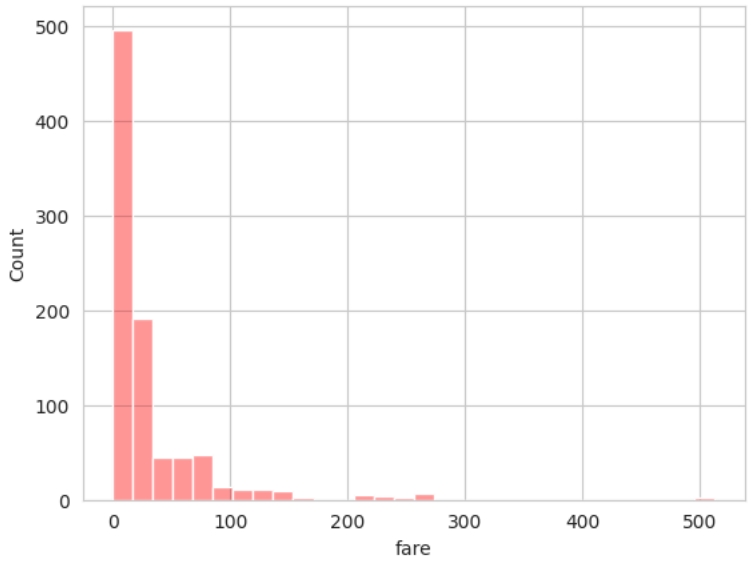
Далее: Создать код с переменной `titanic` New interactive sheet

Q4 plots

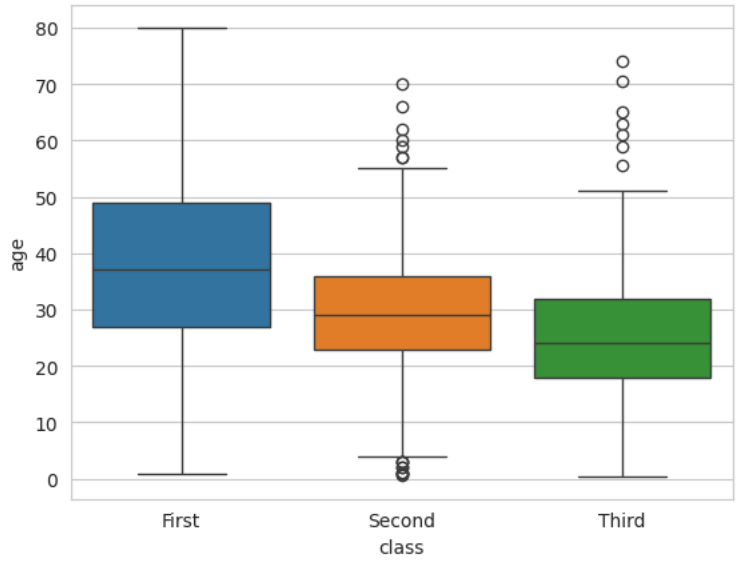
```
sns.jointplot(data=titanic, x='fare', y='age', kind='scatter')
plt.show()
```



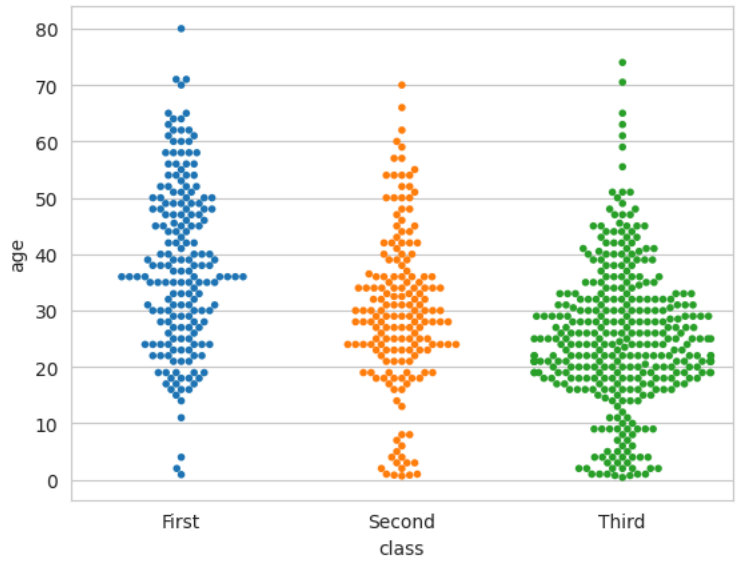
```
sns.histplot(data=titanic, x='fare', bins=30, color='red', alpha=0.4)
plt.show()
```



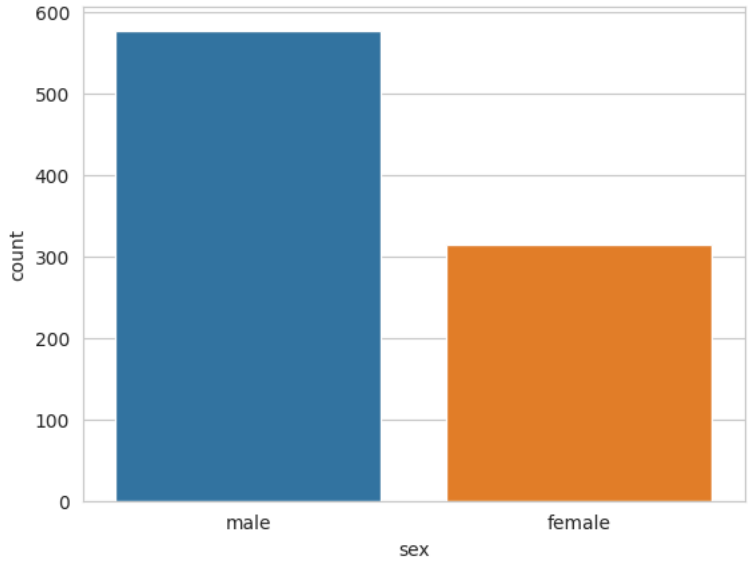

```
sns.boxplot(
    data=titanic,
    x='class',
    y='age',
    hue='class',
    palette=['#1f77b4', '#ff7f0e', '#2ca02c'],
    dodge=False,
    legend=False
)
plt.show()
```




```
sns.swarmplot(
    data=titanic,
    x='class',
    y='age',
    hue='class',
    palette=['#1f77b4', '#ff7f0e', '#2ca02c'],
    size=4,
    legend=False
)
plt.show()
```



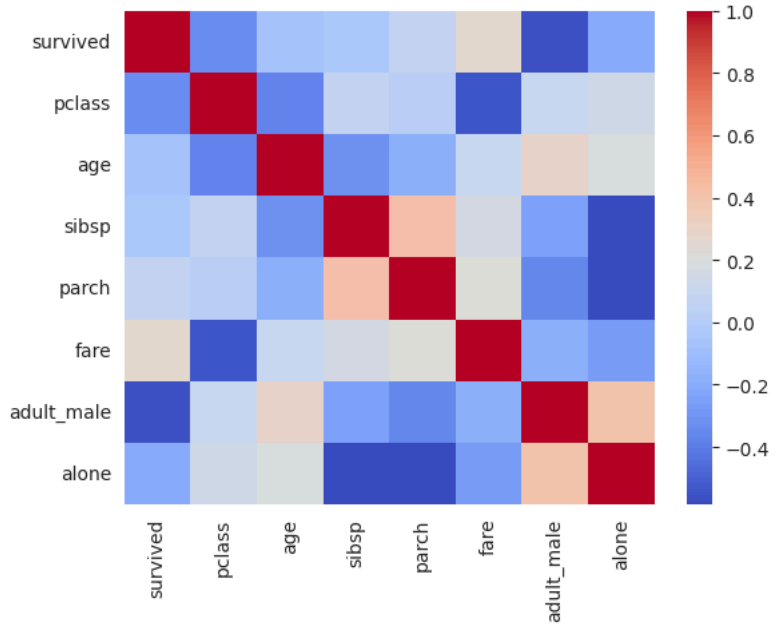

```
sns.countplot(
    data=titanic,
    x='sex',
    hue='sex',
    order=['male', 'female'],
    palette={'male': '#1f77b4', 'female': '#ff7f0e'},
    legend=False
)
plt.show()
```




```
columns = [
    'survived', 'pclass', 'age', 'sibsp',
    'parch', 'fare', 'adult_male', 'alone'
]

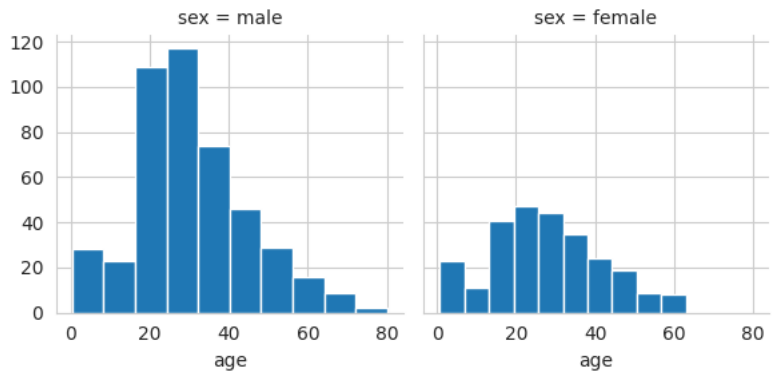
correlation = titanic[columns].corr()

sns.heatmap(correlation, cmap='coolwarm')
plt.show()
```




```
g = sns.FacetGrid(
    titanic,
    col='sex',
    col_order=['male', 'female']
)

g.map(plt.hist, 'age', bins=10, color='#1f77b4', edgecolor='white')
plt.show()
```



Напишите программный код или сгенерируйте его с помощью искусственного интеллекта.